

IT3011 - Theory & Practices in Statistical Modelling



Linear Regression with Multiple Responses



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Linear Regression with A Single Response

The equation for a regression model is as follows,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

Consider the following population dataset,

X_1	X_2	...	X_k	Y
X_{11}	X_{12}	...	X_{1k}	Y_1
X_{21}	X_{22}	...	X_{2k}	Y_2
...	...	...	...	...
X_{n1}	X_{n2}	...	X_{nk}	Y_n

Consider that a regression model is going to be fitted to the above dataset with k independent variables to the response variable Y.

Linear Regression with A Single Response

The equation for a regression model is as follows,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

Consider the i^{th} observation,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \dots + \beta_k X_{ik} + \varepsilon_i$$

Linear Regression with A Single Response

The equation for a regression model is as follows,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

For each observation in the n observations in the population data,

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + \beta_3 X_{13} + \dots + \beta_k X_{1k} + \varepsilon_1$$

$$Y_2 = \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + \beta_3 X_{23} + \dots + \beta_k X_{2k} + \varepsilon_2$$

$$Y_3 = \beta_0 + \beta_1 X_{31} + \beta_2 X_{32} + \beta_3 X_{33} + \dots + \beta_k X_{3k} + \varepsilon_3$$

$\vdots$

$$Y_n = \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + \beta_3 X_{n3} + \dots + \beta_k X_{nk} + \varepsilon_n$$

Matrix Form of Linear Regression with A Single Response

The equation for a regression model is as follows,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

This can be written in following matrix form,

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$



Design Matrix

Matrix Form of Linear Regression with A Single Response

The equation for a regression model is as follows,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

Then this can be written as,

$$Y_{n \times 1} = X_{n \times k+1} \beta_{k+1 \times 1} + \varepsilon_{n \times 1}$$

Matrix Form of Linear Regression with A Single Response

Consider the following example. Consider these are population data.

X	0	1	2	3	4
Y	1	4	3	8	9

What is the design matrix and the response vector.

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 9 \end{bmatrix}$$

Matrix Form of Linear Regression with A Single Response – Dummy Variables

Suppose we need to fit a linear regression model for GPA against the independent variable University Year

GPA (Y) ~ University Year (X: 1st Year, 2nd Year, 3rd Year, 4th Year)

	U_1	U_2	U_3
1 st Year	0	0	0
2 nd Year	1	0	0
3 rd Year	0	1	0
4 th Year	0	0	1

Consider following dataset.

X	1	2	2	3	4
Y	3.5	2.0	2.5	4.0	2.5

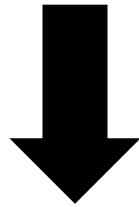
Design Matrix

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Matrix Form of Linear Regression with A Single Response – Least Square Estimation

By minimizing this SSE, we can obtain the parameter estimations as discussed in previous. These estimated parameters can be represented as a vector. The parameter vector β can be obtained through the following matrix operation.

$$\frac{\partial S(\beta)}{\partial \beta_j} = 0 \text{-----} \rightarrow j^{th} \text{ equation}$$



$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Matrix Form of Linear Regression with A Single Response – Least Square Estimation

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

When consider the shape of these matrices X & Y,

$$X_{n \times k+1}$$

$$Y_{n \times 1}$$

Since $X^T_{k+1 \times n}$,

$$(X^T X)_{k+1 \times k+1}$$

$$(X^T X)^{-1} X^T_{k+1 \times n}$$

So,

$$\hat{\beta} = (X^T X)^{-1} X^T Y_{k+1 \times 1}$$

Matrix Form of Linear Regression with A Single Response – Least Square Estimation

So the estimated parameter vector can be shown as follows,

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix}_{k+1 \times 1}$$

Matrix Form of Linear Regression with A Single Response – Predictions & Residuals

Then the estimated response values (predictions) can be obtained by,

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y$$

Still there is a difference between actual response values and the estimated response values. These are called as residuals.

$$\hat{\varepsilon} = Y - \hat{Y}$$

Matrix Form of Linear Regression with A Single Response

Ex:-

Following sample data shows a response variable Y with an independent variable X. By considering a model fitted for these variables, find following things.

X	0	1	2	3	4
Y	1	4	3	8	9

1. Estimated parameter vector
2. Prediction vector
3. Residual vector

Matrix Form of Linear Regression with A Single Response

First following things should be found,

$$\mathbf{X}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 9 \end{bmatrix}$$

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 5 & 10 \\ 5 & 30 \end{bmatrix}$$

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.1 \end{bmatrix}$$

$$\mathbf{X}^T \mathbf{Y} = \begin{bmatrix} 25 \\ 70 \end{bmatrix}$$

Matrix Form of Linear Regression with A Single Response

Estimated parameter vector ($\hat{\beta}$)

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.1 \end{bmatrix} \begin{bmatrix} 25 \\ 70 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Prediction vector ($\hat{Y}$)

$$\hat{Y} = \mathbf{X} \hat{\beta} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$$

Residual vector ($\hat{\varepsilon}$)

$$\hat{\varepsilon} = \mathbf{Y} - \hat{Y} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 9 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

Consider there are m response variables,

$$Y_1, Y_2, Y_3, \dots, Y_m$$

and a single set of independent variables,

$$X_1, X_2, X_3, \dots, X_k$$

Each variable is assumed to follow its own regression model. so that,

$$Y_1 = \beta_{01} + \beta_{11}X_1 + \dots + \beta_{k1}X_k + \varepsilon_1$$

$$Y_2 = \beta_{02} + \beta_{12}X_1 + \dots + \beta_{k2}X_k + \varepsilon_2$$

$$\vdots$$

$$Y_m = \beta_{0m} + \beta_{1m}X_1 + \dots + \beta_{km}X_k + \varepsilon_m$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

The design matrix is same as previous

$$X_{n \times k+1} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}$$

Responses, parameters and errors are in the matrix form now.

$$Y_{n \times m} = [Y_{(1)} : Y_{(2)} : \dots : Y_{(m)}] \quad \beta_{k+1 \times m} = [\beta_{(1)} : \beta_{(2)} : \dots : \beta_{(m)}] \quad \varepsilon_{n \times m} = [\varepsilon_{(1)} : \varepsilon_{(2)} : \dots : \varepsilon_{(m)}]$$
$$Y_{n \times m} = \begin{bmatrix} Y_{11} & Y_{12} & \dots & Y_{1m} \\ Y_{21} & Y_{22} & \dots & Y_{2m} \\ Y_{31} & Y_{32} & \dots & Y_{3m} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n1} & Y_{n2} & \dots & Y_{nm} \end{bmatrix} \quad \beta_{k+1 \times m} = \begin{bmatrix} \beta_{01} & \beta_{02} & \dots & \beta_{0m} \\ \beta_{11} & \beta_{12} & \dots & \beta_{1m} \\ \beta_{21} & \beta_{22} & \dots & \beta_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{k1} & \beta_{k2} & \dots & \beta_{km} \end{bmatrix} \quad \varepsilon_{n \times m} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \dots & \varepsilon_{1m} \\ \varepsilon_{21} & \varepsilon_{22} & \dots & \varepsilon_{2m} \\ \varepsilon_{31} & \varepsilon_{32} & \dots & \varepsilon_{3m} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{n1} & \varepsilon_{n2} & \dots & \varepsilon_{nm} \end{bmatrix}$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

Then this can be written as,

$$Y_{n \times m} = X_{n \times k+1} \beta_{k+1 \times m} + \varepsilon_{n \times m}$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses) - Least Square Estimation

Simply the i^{th} response $Y_{(i)}$ follows the linear regression model,

$$Y_{(i)} = \mathbf{X}\beta_{(i)} + \varepsilon_{(i)}$$

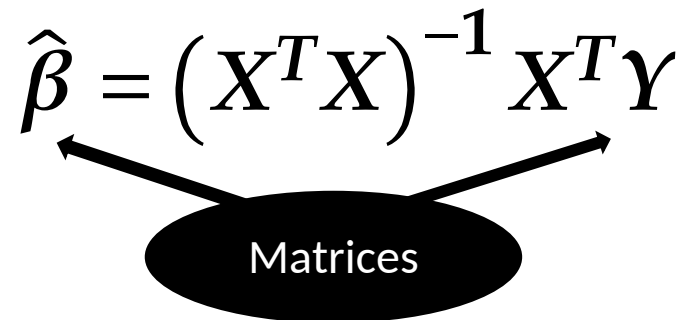
where $i=1,2,\dots,m$

Then its estimated parameter vector can be shown as,

$$\hat{\beta}_{(i)} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y_{(i)}$$

So for the entire set of parameters,

$$\hat{\boldsymbol{\beta}} = [\hat{\beta}_{(1)} : \hat{\beta}_{(2)} : \dots : \hat{\beta}_{(m)}] = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T [Y_{(1)} : Y_{(2)} : \dots : Y_{(m)}]$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$


Matrices

Multivariate Multiple Regression (Linear Regression with Multiple Responses) – Predictions & Residuals

With the estimated parameter matrix, following results can be obtained.

$$\hat{Y} = X\hat{\beta}$$

$$\hat{\varepsilon} = Y - \hat{Y}$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

Ex:-

Following sample data shows two response variables Y_1 and Y_2 with an independent variable X . By considering a set of models fitted for these variables, find following things.

X	0	1	2	3	4
Y_1	1	4	3	8	9
Y_2	-1	-1	2	3	2

1. Estimated parameter matrix
2. Prediction matrix
3. Residual matrix

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

First following things should be found,

$$\mathbf{X}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & -1 \\ 4 & -1 \\ 3 & 2 \\ 8 & 3 \\ 9 & 2 \end{bmatrix}$$

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.1 \end{bmatrix}$$

Multivariate Multiple Regression (Linear Regression with Multiple Responses)

Estimated parameter matrix ($\hat{\beta}$)

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

Prediction matrix ($\hat{Y}$)

$$\hat{Y} = X\hat{\beta} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 5 & 1 \\ 7 & 2 \\ 9 & 3 \end{bmatrix}$$

Residual matrix ($\hat{\varepsilon}$)

$$\hat{\varepsilon} = Y - \hat{Y} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ -2 & 1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$$